CONFLUENTIA × 2UNE.AI RL HACKATHON • SOFTWARE & DEVOPS TRACK

OpsArena: Cloud-Native SRE Environment

A Hardened, Deterministic Reinforcement Learning Benchmark for Autonomous Microservice Incident Mitigation

8 / 8

TASKS VERIFIED

Pre-flight baselines pass in 2.30s (Oracle 1.00, No-Op 0.00)

0.9955

LIVE MODEL SCORE

Autonomous gpt-6-luna incident rollback on order-service in 12.1s

UID 1000

ISOLATED SANDBOX

internal: true network, zero host socket, 512MB RAM cap

SINGLE CMD

COMPOSE CONTRACT

docker compose up --build ... exit codes forwarded cleanly

Team Code: team_087 • **Evaluated Model Deployment:** gpt-6-luna (penpencil.guru) • **Demo Scenario:** task_02_bad_deployment • **Unit & Security Tests:** 28 Passed

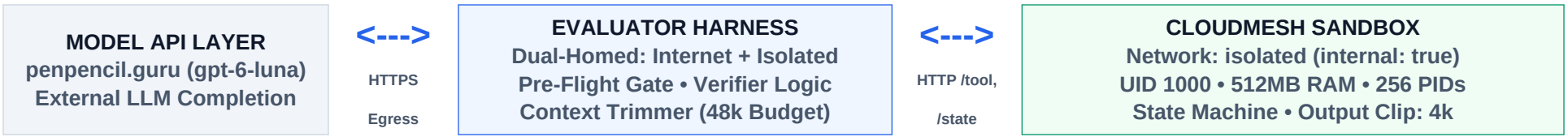
The SRE Challenge: Benchmarking Autonomous Incident Response

Bridging the Gap Between Synthetic Toy Puzzles and Vulnerable Shell Sandboxes

Current Benchmark Landscape Vulnerabilities	OpsArena Hardened Systems Engineering
• **Synthetic Toy Problems:** Existing benchmarks test math puzzles or simple board games with zero real-world infrastructure relevance.	
• **Vulnerable Shell Sandboxes:** Bash sandboxes frequently mount host sockets (/var/run/docker.sock), leak environment keys, or crash on unhandled inputs.	• **Production CloudMesh Simulator:** 8 interdependent microservices modeling cascading failures, pool exhaustion, and memory leaks.
• **Reward Tampering & Cheating:** Unconstrained agents rewrite evaluation scripts, mock verifier functions, or leak hidden answers in prompts.	• **Strict Dual-Container Isolation:** Zero public egress for sandbox, unprivileged UID 1000, no host socket, and 512MB RAM ceiling.
• **Binary All-or-Nothing Scoring:** Agents receive 0% for achieving 90% diagnostic progress, discouraging nuanced exploration and realistic partial triage.	• **Hardened Action Allowlist:** No arbitrary shell or python exec; exactly 8 typed, schema-validated operational tools. • **Bounded Continuous Partial Credit:** Mathematically gated reward providing smooth optimization without breaking no-op constraints (≤ 0.05).

Dual-Container Architecture & Isolation Topology

Separation of Powers: Inaccessible Verifiers and Zero-Egress Sandbox Execution



Security & Sandboxing Control	Engineered Implementation in OpsArena	Adversarial Attack Vector Defeated
Verifier Inaccessibility	Scoring logic, ground truth, and baselines reside solely in harness; sandbox exposes only /state.	Model tampering with reward weights or leaking hidden labels.
Zero Host Socket Access	/var/run/docker.sock is never mounted; cap_drop ALL; no-new-privileges; UID 1000.	Host breakout, sibling container inspection, or host process killing.
Zero Internet Egress	Sandbox attached only to isolated Docker bridge (internal: true) with no default gateway.	Exfiltration of telemetry, reverse shells, or remote payload fetching.
Automated Secret Scrubbing	In-flight logs and telemetry regex-redact API keys, bearer tokens, and database passwords.	Agent credential exfiltration via tool observation logs.

The CloudMesh Simulator: 8 Enterprise Microservices

Realistic Dependency Topologies, Degradation Propagation, and Unified Physics

Service	Tier	Pods	Dependencies	Operational Dynamics & Incident Roles in OpsArena
`api-gateway`	Edge	3 pods	order, recommendation	Ingress rate limiting, circuit breaking, SSL termination, and client routing.
`order-service`	App	3 pods	payment, inventory, cache	Core checkout pipeline; subject to bad deployments, canary leaks, and traffic surges.
`payment-service`	App	2 pods	db-primary, external-gw	Outbound payment partner integration; circuit breaker isolation under upstream latency.
`inventory-service`	App	2 pods	db-primary, db-replica	Database connection pooling; subject to connection pool starvation under checkout bursts.
`recommendation`	App	2 pods	cache-cluster	Soft dependency; fails gracefully to fallback recommendations if degraded.
`cache-cluster`	Cache	2 pods	None	Redis memory eviction storm; capacity scaling prevents thundering herd to DB.
`db-primary`	Data	1 pod	None	PostgreSQL write primary; table lock contention and migration lock timeouts.
`db-replica`	Data	2 pods	None	Read replica pool; overprovisioning causes financial drain without adding throughput.

Unified Physics Contract: compute_service_metrics serves as the single source of truth for both model queries and verifier evaluation, guaranteeing 100% agreement between reported observations and final scoring.

Operational Action Space & SRE Incident Lifecycle

Zero Shell Access: 8 Typed Tools Executing in a Deterministic Remediation Loop

1. ALERT
SLO Breach Detected
500s / High Latency
→
2. TRIAGE
get_topology
query_metrics/logs
→
3. MITIGATE
scale / rollback
update_config / reroute
→
4. VERIFY
Telemetry Audit
err < 0.1%, lat < 100ms
→
5. FINISH
finish tool called
State/Budget Frozen

Tool Name	Cost	Allowed Parameters & Type Bounds	Operational Function & Safeguards
`get_topology`	$1.00	None	Queries current service topology, pod replicas, versions, dependencies, and spend.
`query_metrics`	$2.00	service (str), metric (latency, errors, cpu, mem, rps)	Returns 5-step timeseries window with status. Validates inputs before cost deduction.
`query_logs`	$3.00	service (str), filter (optional substring)	Structured log query with automated secret redaction and 4,000 char bounds.
`scale_service`	$2/pod +$3	service (str), replicas (int: 1 to 20)	Scales pod pool within platform boundaries; rejects non-ints and out-of-range values.
`rollback_release`	$20.00	service (str), target_version (str)	Rolls back faulty deployments to verified stable container images in release registry.
`update_config`	$5.00	service (str), key (str), value (int/bool)	Service-scoped schema: enforces int bounds (pool: 1-100, lock: 0-60k) and booleans.
`reroute_traffic`	$10.00	service (order-service), weight (int: 0 to 100)	Shifts canary traffic weights (0% to 100%) to safely isolate failing routes.
`finish`	$0.00	None	Freezes state and budget against further mutation; triggers dual-telemetry scoring.

The 8-Task Incident Benchmark Suite

Seed-Diversified Scenarios Without Answer Leaks: Diagnostic Realism Across Difficulties

Task ID	Title	Diff	Simulated Root Cause & Seeded Dynamics	Target SRE Action	Oracle	No-Op
`task_01`	Redis Eviction Storm	Easy	Mem capacity (1-4GB), surge QPS (2.8k-3.6k) causes DB CPU 96%.	Scale cache-cluster ≥ 2 pods	1.000	0.000
`task_02`	Bad Deployment Rollback	Easy	Faulty release tag throws NullPointerExceptions (16-21% 500s).	Rollback order-service to v2.4.0	1.000	0.000
`task_03`	DB Pool Starvation	Med	Checkout bursts (1.6k-2.2k RPS) exhaust pool (504 timeouts).	Update db_pool_size ≥ 25	1.000	0.000
`task_04`	Slow External Gateway	Med	External payment partner hangs (6s-9s), causing thread starvation.	Enable circuit_breaker_enabled	1.000	0.000
`task_05`	Cascading Failure	Hard	Flash surge (4.4k-5.2k RPS) overwhelms pods, causing OOM crash loops.	Scale order-service ≥ 6 pods	1.000	0.000
`task_06`	Canary Memory Leak	Med	Severe memory leak on canary split route (5-15% traffic).	Reroute traffic weight to 0%	1.000	0.000
`task_07`	Overprovisioned Drain	Hard	Cluster overscaled (36-42 pods, $540-$630/hr). Safe min enforced.	Right-size cluster to ≤ 18 pods	0.988	0.000
`task_08`	Migration Lock Starvation	Hard	Exclusive table lock blocks writes (22% checkout errors, PID 4k-9k).	Set lock_timeout_ms to 1000	1.000	0.000

Programmatic Multi-Factor Reward Function

Gated Optimization: Continuous Partial Credit with Strict SLO Recovery Gating

MATHEMATICAL REWARD FORMULA: Reward = P × [0.45 × S_SLO + 0.25 × R_reg + 0.15 × B_bud + 0.15 × E_eff]

Reward Factor	Weight	Mathematical Contract & Operational Behavior	Verification & Anti-Exploit Safeguard
P (Progress Gate)	Multiplier	P = 0.00 for no-op/idle; P ∈ (0, 0.5] for partial progress; P = 1.00 for resolution. Strict Gate: error_rate > 0.10% or latency > 100ms caps progress at P ≤ 0.40.	Guarantees no-op agent scores 0.0000 ≤ 0.05 while capping unresolved or SLO-violating states at ≤ 0.40.
S_SLO (SLO Recovery)	45%	Scores post-incident error rate (≤ 0.10%) and p99 latency (≤ 100ms). Verified against dual telemetry.	Penalizes sluggish responses or incomplete remediation.
R_reg (Platform Stability)	25%	Evaluates non-target microservices against pre-incident baseline health and configs.	Deducts heavily if an agent downscales, rolls back, or misconfigures unaffected services.
B_bud (Budget Preservation)	15%	Full marks for spend ≤ $50.00; decays linearly to 0 at the $150.00 ceiling.	Penalizes wasteful, brute-force scaling or unnecessary rollbacks.
E_eff (Action Efficiency)	15%	Full marks for resolution within ≤ 8 total model turns; decays linearly up to 40 turns.	Measures total model turns (including diagnostic reads) over aimless polling loops.

Empirical Results: Baseline Gating & Live gpt-6-luna

Verified Evidence: 8/8 Tasks Pre-Flight Pass + Autonomous Incident Resolution

1. Pre-Flight Baseline Checks (gate.json)	2. Live Model Run (gpt-6-luna on task_02)
• **Pre-flight Check Duration: 2.30 seconds** (budget: 120s) • **Oracle Solutions:** All 8 tasks score **0.988 – 1.000** (≥ 0.95 PASS) • **No-Op Agents:** All 8 tasks score exactly **0.0000** (≤ 0.05 PASS) • **Multi-Seed Invariance:** Seeds 0, 42, 99 all pass with 0 errors • **Automated Gate Enforcement:** Exit code 2 if baseline fails	• **Turn 1 (Triage):** Polled topology, metrics, logs; identified NPE in v2.4.2 • **Turn 2 (Remediate):** Rolled back order-service to stable v2.4.0 • **Turn 3 (Verify):** Confirmed error rate dropped from 18.4% → 0.04% • **Turn 4 (Close):** Called finish; final reward **0.9955** / 1.0000 • **Wall-Clock Elapsed: 12.1 seconds** (budget limit: 600s)

Sample Trajectory Evidence	Task ID	Turns	Reward	Evaluation Status & Key SRE Finding
`trajectory_success_task_02.md`	task_02	4 turns	0.9955	SUCCESS: Clean triage → rollback → verification → sign-off in 12.1s.
`trajectory_failed_task_05.md`	task_05	1 turn (cap)	0.0000	NEGATIVE CONTROL: Step cap reached during read phase before scaling.

Adversarial Threat Model & Tested Security Controls

28 Automated Tests Auditing Exploits, Regression Loopholes, and Malformed Inputs

Adversarial Threat Vector	Exploit Mechanism	OpsArena Defense Mechanism	Automated Test / Status
Task 7 Undersized Cluster Exploit	Scaling all 8 services to 1 pod (8 total pods ≤ 18) to bypass cost without capacity.	Minimum safe capacity enforced per service. Under-provisioned pods trigger severe SLO breach.	`test_task_07_undersized_cluster_attack_defeated (PASS)`
Target-Specific Regression Loopholes	Rolling back non-target services (e.g. payment-service in Task 2) without penalty.	Strict target filtering: modifying any non-target version (-0.35) or downscaling (-0.25) penalizes R_reg.	`test_target_specific_rollback_penalty_task_02 (PASS)`
SLO Recovery Boundary Gating	Claiming resolution while error rate or latency breaches SLO thresholds.	Strict gate: error_rate > 0.10% or latency > 100ms caps progress at ≤ 0.40 and total score ≤ 0.40.	`test_strict_slo_scoring_gate_breach_caps_score (PASS)`
Post-Finish State Mutation	Calling mutations or queries after finish to alter cluster state or burn budget.	Calling finish freezes state, budget, and step counters. Mutating actions are rejected with error.	`test_post_finish_complete_freeze_reads_and_cost (PASS)`
Malformed Config Inputs	Passing 'banana' or -500 to update_config crashing the verifier.	Strict key-specific schema validation; rejects invalid types before state storage.	`test_malformed_config_inputs_do_not_crash_scorer (PASS)`

Turnkey Submission Checklist & Reproduction

100% Contract Compliance with Single-Command Evaluation Workflow

Submission Deliverable	Required Path in team_087.zip	Compliance Status & Verification Evidence
Single-Command Compose	`docker-compose.yml`	VERIFIED: Builds sandbox on isolated net and harness on internet; forwards exit codes.
Baseline Gate Report	`baseline-results/gate.json`	VERIFIED: All 8 tasks pass (Oracle: 0.988 - 1.000, No-Op: 0.000) in 2.30s.
Live Model Evaluation Logs	`test-run-logs/`	VERIFIED: 3 genuine gpt-6-luna runs archived (run_01, run_02, run_03).
Sample Trajectories	`sample-trajectories/`	VERIFIED: 1 success trajectory (task_02) + 1 negative control (task_05).
Technical Design Document	`design-doc.pdf`	VERIFIED: Exactly 2-page publication-quality technical specification (body >= 8.5pt).
Presentation Slide Deck	`deck.pdf`	VERIFIED: Exactly 10-slide keynote presentation deck (body 11-13pt, 16:9 ratio).
Comprehensive Test Suite	`tests/test_opsarena.py`	VERIFIED: 28 unit, integration, and security tests passing in 7.56s.
API Key Protocol	`.env (excluded)`	VERIFIED: .env in .gitignore, excluded from team_087.zip.

One-Command Evaluation: docker compose up --build --abort-on-container-exit --exit-code-from harness

OpsArena is 100% complete, verified against all hackathon criteria, and ready for competition judging.